RESEARCH PAPER

FACTORY REALLOCATION & SHIPPING OPTIMIZATION

A Scenario-Based Decision Support System for Nassau Candy Distributor

Project: Unified Mentor Internship Project

Dataset: 10,194 order records • 15 products • 5 factories

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Abstract

This paper describes a data-driven decision support system designed to examine factory assignment and shipping-efficiency opportunities in a confectionery distribution network. The project combines exploratory data analysis, data-quality checks, geographic distance calculations, scenario-based transit-time estimation, machine-learning model comparison, a recommendation-scoring framework, clustering, and an interactive Streamlit dashboard. The dataset contains 10,194 order records and 18 source fields covering orders, dates, shipping mode, customer geography, products, sales, units, gross profit, and cost.

A key methodological distinction is that the source date fields produce implausibly long order-to-ship intervals. Consequently, the project does not treat those raw intervals as trustworthy real-world delivery performance. Instead, the optimization workflow uses an explicitly scenario-based transit-time estimate derived from approximate factory-to-destination distance and stated shipping-mode assumptions. A Random Forest model achieved an R^2 of 0.946, MAE of 0.4899, and RMSE of 0.6685 when evaluated against this generated scenario target. These scores describe agreement with the synthetic target—not accuracy against verified carrier delivery records. The resulting dashboard helps users compare alternatives, understand trade-offs, and shortlist scenarios for operational validation.

The system is a prototype for decision support rather than an autonomous factory-allocation engine. Recommendations require validation against manufacturing capability, capacity, actual freight rates, service commitments, inventory, and product-specific production constraints before implementation.

1. Introduction

Distribution networks must balance service speed, transport distance, manufacturing economics, and customer demand. A fixed product-to-factory assignment can be easy to administer, but it may not always be the best choice for every destination or operating condition. Data analysis can help identify alternatives worth investigating, especially when decision-makers need to compare the potential effect of moving a product from its currently assigned factory to another facility.

This project uses the Nassau Candy Distributor order dataset to build a practical analysis and visualization workflow. The aim is not to claim that a simulated assignment is automatically operationally optimal. Rather, the system organizes available evidence, makes assumptions visible, and provides a repeatable way to screen factory-product alternatives before a business decision is made.

2. Problem Statement and Objectives

The project addresses the lack of a unified analytical view for comparing factory assignments using order economics, geographic proximity, and estimated shipping effort. Without a structured comparison, a team may find it difficult to distinguish a promising alternative from one that reduces distance but damages profitability or is infeasible to manufacture.

The objectives are: (1) inspect and clean the order dataset; (2) derive analysis-ready fields for margin, unit economics, time, and geography; (3) map products to their current factories; (4) estimate factory-to-destination distances using available geographic coordinates; (5) construct a transparent scenario transit-time measure; (6) compare machine-learning regressors against that scenario target; (7) score and rank alternative factory assignments; (8) group similar scenarios through clustering; and (9) deliver the outputs through a Streamlit dashboard and downloadable tables.

3. Dataset and Scope

The working dataset contains 10,194 rows and 18 original columns. The fields cover row and order identifiers, order and ship dates, ship mode, customer identifier, country/region, city, state/province, postal code, division, region, product identifier and name, sales, units, gross profit, and cost. The data includes US and Canadian destinations and 15 distinct products.

The analysis scope is limited to the variables available in the supplied file and the factory/product mapping used in the notebook. The dataset does not provide a complete record of factory capacity, production line compatibility, carrier-level performance, live freight tariffs, inventory constraints, or verified actual delivery timestamps. Those missing operational inputs limit the conclusions that can be drawn from a geographic and financial simulation.

4. Data Preparation and Quality Assurance

The notebook imports the source workbook into pandas, inspects dimensions and data types, checks missing values and duplicates, standardizes date fields, and validates numeric business measures. Derived columns include lead-time-related fields, profit margin, profit/revenue/cost per unit, cost ratio, and order year/month. Invalid or inconsistent date values are treated as a data-quality issue rather than silently interpreted as reliable service outcomes.

A critical limitation emerged in the source dates: the resulting order-to-ship intervals are implausibly long for ordinary shipping. Therefore, these intervals should not be used as verified carrier transit times or as direct proof of operational delays. Any chart or metric based on those dates must be clearly labeled as a source-data diagnostic. The optimization model instead uses a separate synthetic scenario target described in Section 6.

Quality checks performed in the workflow include duplicate-row inspection, null-value review, numeric-value validation, date-range inspection, factory mapping coverage, and geographic coordinate completeness. State/province coordinates represent approximate destination centroids, not precise customer addresses. Distance estimates should therefore be interpreted as directional approximations.

5. Factory Mapping and Geographic Distance

Five factories are included in the analysis: Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory. Product-to-factory mappings provide the current assignment used as the baseline. Factory coordinates are used to estimate the distance from a factory to a destination state's or province's representative coordinate.

For each order or destination grouping, the workflow calculates the distance from the currently assigned factory and compares it with alternative factories. Distance reduction is computed as (current distance − alternative distance) divided by current distance, multiplied by 100, when the baseline distance is positive. This measure helps identify geographic opportunities; it does not account for road-network routing, traffic, cross-border procedures, carrier hubs, temperature control, or actual freight prices.

The current factory mapping is treated as a project input, not as evidence that every factory is technically capable of manufacturing every product. A real implementation would need a product-factory eligibility matrix approved by manufacturing and quality teams.

6. Scenario-Based Transit-Time Estimation

Because the original date-derived intervals are not reliable enough to serve as actual shipping labels, the project creates a transparent scenario target using distance and assumed ship-mode speeds. The calculation is: scenario transit days = mode-specific base days + estimated distance in kilometres divided by the assumed kilometres per day. The assumptions below are project parameters selected to support comparison, not measured carrier service levels.

Same Day uses a base of 1 day and 1,200 km/day; First Class uses 2 days and 900 km/day; Second Class uses 3 days and 700 km/day; and Standard Class uses 5 days and 500 km/day. The labels are retained as scenario categories, but the parameters should not be interpreted as industry benchmarks. Different assumptions can change the ranking of alternatives, so the dashboard should be used for sensitivity analysis rather than as a guarantee of delivery date.

The advantage of this approach is transparency: users can trace how distance and shipping-mode assumptions affect estimated transit time. Its weakness is that the output inherits the limitations of the assumptions. Real carrier data, dispatch timestamps, warehouse processing times, and service-level definitions would be needed to calibrate and validate a real transit-time model.

Ship mode	Base days	Assumed km/day
Same Day	1	1,200
First Class	2	900
Second Class	3	700
Standard Class	5	500

Table 1. Scenario parameters used by the prototype; values are assumptions, not measured carrier standards.

7. Feature Engineering and Exploratory Analysis

Exploratory analysis examines product mix, factory assignments, sales, gross profit, cost, units, margins, destination geography, and shipping modes. Engineered measures such as profit margin, cost ratio, per-unit economics, order month, and distance provide additional ways to compare products and potential reassignment scenarios.

Profit margin is calculated as gross profit divided by sales, multiplied by 100, when sales are non-zero. Unit-level measures divide sales, gross profit, or cost by units when units are positive. These metrics are descriptive of the supplied records and should be interpreted in the context of the dataset's definitions and completeness.

Visualizations are used to compare factory and product profiles, inspect the distribution of distances and scenario transit times, and highlight candidate alternatives. Descriptive patterns are useful for screening, but they do not establish causality. For example, a high margin observed at one factory does not prove that a product would retain the same margin if transferred there; manufacturing costs and production conditions could differ by product and site.

8. Machine-Learning Methodology

Three regression approaches—Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor—were compared for predicting the scenario-based transit-time target. The model workflow separates the predictors and target, applies a train/test evaluation, and compares prediction errors. The Random Forest model achieved RMSE 0.6685, MAE 0.4899, and R^2 0.946 in the recorded experiment.

RMSE measures the square root of average squared prediction error, so larger errors receive greater weight. MAE reports the average absolute prediction error in the target's units. R^2 measures the proportion of target variance explained relative to a baseline mean prediction. These values must be read with the target definition in mind: because the target is generated from an explicit formula, a high score indicates that the model approximates that formula within the evaluation split. It does not demonstrate equivalent accuracy against actual delivery outcomes.

The model comparison is useful as a technical exercise and can support a future model pipeline, but the scenario formula itself may be simpler and more interpretable than a learned prediction. For production use, a future study should compare models against verified historical transit-time labels, use leakage-aware validation, report residual plots and segment-level errors, and test generalization across time periods and destinations.

Model	RMSE	MAE	R^2
Linear Regression	**1.8041**	**1.5402**	**0.6068**
Random Forest	**0.6685**	**0.4899**	**0.9460**

Regressor			
Gradient Boosting Regressor	**0.8682**	**0.6853**	**0.9089**

Table 2. Model Comparison. Performance comparison of Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor using RMSE, MAE, and R^2. Random Forest achieved the lowest RMSE and MAE and the highest R^2 among the three evaluated models.

9. Recommendation Engine and Scoring

The recommendation engine evaluates alternative factories for products and estimates how distance, scenario transit time, margin, and confidence may change under each scenario. It produces fields such as current factory, alternative factory, current and alternative distance, distance reduction, estimated transit-time reduction, margin, units or sales context, confidence, and recommendation score.

The project score is a policy-weighted decision aid, not a model probability. The current framework uses four components: estimated lead-time reduction (35%), distance reduction (30%), profit margin (20%), and scenario confidence (15%). These weights are chosen for the prototype and can be adjusted after stakeholder review. Before combining them, component values should be normalized to comparable scales so one variable does not dominate merely because of its numeric range.

A high score means a scenario ranks well under the chosen assumptions and weights. It does not mean the recommendation is guaranteed to save money or is approved for execution. A complete financial model would need actual transport rates, transfer costs, manufacturing cost changes, taxes or duties where applicable, service penalties, and capacity costs.

10. Clustering and Scenario Segmentation

K-Means clustering is used to group scenarios with similar characteristics, using variables such as distance, scenario transit time, estimated savings or reductions, and profit margin. Four clusters are used in the current project configuration. The intent is to help users compare broad scenario patterns and explore groups of candidates rather than inspect each record in isolation.

Clustering is sensitive to feature scaling, outliers, the selected number of clusters, and random initialization. Numeric features should be standardized before fitting when they use different units. Cluster labels are identifiers, not ranked business categories; each cluster must be interpreted by examining its profile and representative scenarios. The cluster output is exploratory and should not be presented as proof that a factory reassignment is viable.

11. Illustrative Recommendation Example

One illustrative scenario in the project is reassignment of Wonka Bar – Triple Dazzle Caramel from Wicked Choccy's to Lot's O' Nuts. The scenario calculations show an estimated current distance of approximately 2,828.56 km and an alternative distance of approximately 99.34 km, corresponding to an estimated distance reduction of about 96.49%. The recorded scenario also shows an estimated transit-time reduction of about 67.75%, a product margin of approximately 65.33%, and a recommendation score near 79.22.

These numbers are outputs of the project's scenario assumptions and data mapping. They are not observed savings, actual route measurements, or proof that Lot's O' Nuts can produce and ship the product. The large distance difference makes this scenario useful for demonstrating the dashboard, but it should be validated against factory eligibility, customer demand, capacity, routing, freight cost, and service constraints before any operational decision.

Metric	Scenario value
Current factory	Wicked Choccy's
Alternative factory	Lot's O' Nuts
Current estimated distance	2,828.56 km
Alternative estimated distance	99.34 km
Estimated distance reduction	96.49%
Scenario transit-time reduction	67.75%
Product margin	65.33%
Recommendation score	79.22

Table 3. Illustrative scenario output; values require operational validation.

12. Streamlit Decision-Support Dashboard

The Streamlit application provides a visual interface for exploring the analysis without requiring users to rerun every notebook cell. The current dashboard includes Home, AI Decision Center, Factory Optimization, What-If Analysis, Recommendations, Risk & Impact, Data Insights, ML Performance, Route Clustering, and Reports & Export pages.

The dashboard supports high-level KPI review, factory and product filtering, side-by-side scenario comparison, recommendation ranking, risk review, model metric inspection, cluster exploration, and export of analytical tables. The interface is intended for exploratory decision support: it makes the analysis easier to review, but the quality of any decision remains dependent on source data and assumptions.

The supporting project outputs include final factory recommendations, product recommendations, route-cluster analysis, KPI summary, and model-comparison tables. The notebook provides the reproducible analysis workflow, while the Streamlit application presents the outputs in a form suitable for a project demonstration.

13. Results and Discussion

The project demonstrates an end-to-end analytical workflow from workbook inspection to an interactive recommendation dashboard. It combines business-oriented measures with geographic comparison and a clearly defined scenario target. The recorded Random Forest metrics show that the chosen model can reproduce the generated target reasonably well in the evaluated split; however, the synthetic nature of that target limits the interpretation of the model's apparent accuracy.

The recommendation ranking makes competing objectives visible. An alternative may be attractive because it is geographically closer, but the same option may be weak on margin or operational feasibility. Separating the score into components allows reviewers to challenge the assumptions, change weights, and identify which factor is responsible for a high or low ranking.

The most valuable outcome of the prototype is therefore the structured comparison process rather than a claim of realized cost savings. It gives a reviewer a repeatable way to screen alternatives, document assumptions, and define the data required for a proper pilot.

14. Limitations, Risks, and Ethical Use

The main limitation is the absence of validated actual shipping outcomes suitable for supervised learning. The original date fields create implausibly long intervals, and the scenario target is formula-generated. Model metrics must not be represented as verified delivery-time accuracy.

Other limitations include approximate state/province centroid coordinates; absence of road-network distances and carrier routing; missing factory capacity and production compatibility data; lack of actual freight rates and transfer costs; potentially incomplete information about seasonal demand and inventory; and the subjective nature of recommendation weights. A geographic alternative may also be unsuitable due to quality certification, equipment, recipe, packaging, or regulatory requirements.

The dashboard should be used as a prototype for review, not as an automated system that executes factory changes. Before publishing or sharing project files, customer identifiers and location fields should be reviewed for privacy and authorization. Any public repository should contain only data that the user is permitted to disclose, or a suitably anonymized sample.

15. Recommendations for Future Work

First, obtain validated timestamps for order acceptance, production completion, dispatch, carrier pickup, and delivery. Second, add factory capacity, eligible product lines, inventory, service-level targets, and carrier-level transit data. Third, calculate actual freight costs and include fixed transfer, setup, packaging, and quality-validation costs. Fourth, evaluate alternatives with time-based validation and report error distributions across regions, ship modes, and products.

A future optimization engine could formulate factory assignment as a constrained optimization problem with binary decision variables, demand and capacity constraints, product-factory eligibility, service-level requirements, and a cost objective. Scenario analysis should include sensitivity to weight choices, fuel prices, volume changes, and disruption risks. A pilot should track actual outcomes against a baseline before any recommendations are scaled.

16. Conclusion

This project presents a prototype Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor. It combines data preparation, exploratory analysis, factory-distance comparison, scenario-based transit-time estimation, regression model evaluation, weighted recommendation scoring, clustering, and an interactive Streamlit dashboard. The system helps users compare alternatives and understand the trade-offs among geographic distance, estimated transit time, margin, and confidence.

The findings are best interpreted as hypotheses for further operational validation. Because the target transit times are scenario-generated and several real-world constraints are unavailable, the project does not claim verified delivery-time improvements or realized financial savings. With reliable logistics labels, manufacturing constraints, and freight-cost data, the prototype could be extended into a more defensible operational planning tool.

References

McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.

Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Streamlit. Streamlit documentation. https://docs.streamlit.io/

Scikit-learn. User Guide and API documentation. https://scikit-learn.org/stable/user_guide.html

Project source: Nassau Candy Distributor order dataset supplied for the Unified Mentor project; analysis notebook, exported recommendation tables, and Streamlit application developed as part of this project.

Appendix A. Key Definitions

Estimated distance reduction (%) = (current estimated distance − alternative estimated distance) / current estimated distance × 100, when current distance is greater than zero.

Profit margin (%) = gross profit / sales × 100, when sales is greater than zero.

Scenario transit days = ship-mode base days + estimated distance in kilometres / assumed kilometres per day.

MAE = mean of absolute differences between predicted and reference target values. RMSE = square root of the mean squared prediction error. R^2 = coefficient of determination relative to a mean-prediction baseline.

Recommendation score = weighted combination of normalized scenario indicators. In this project the weights are 35% scenario transit-time reduction, 30% distance reduction, 20% profit margin, and 15% scenario confidence.

Appendix B. Reproducibility Checklist

Before submission, verify that the notebook runs from the first cell to the final export cell; the input dataset path is correct; factory mapping and coordinates are documented; scenario parameters are visible in the notebook; model metrics match the current notebook output; exported CSVs correspond to the final code; the Streamlit dashboard loads those outputs correctly; and the report clearly labels all scenario-derived metrics.

Before operational use, replace synthetic transit-time targets with verified historical labels, validate the mapping with business owners, review privacy permissions for all published files, and obtain approval from manufacturing, logistics, finance, and quality teams.